

Multi-Objective Pareto Hyperparameter Optimization for Joint Detection and Localization of Forged Identity Documents

Hernán Díaz Rodríguez^{a,*}

^aDepartment of Computer Science, University of Oviedo, Asturias, Spain

Abstract

Automated forensic analysis of identity documents requires simultaneously detecting whether a document has been tampered with and identifying which regions were altered. These two objectives are structurally coupled but exist in tension: optimizing one with a fixed loss weight may degrade the other. We propose DocVerify, a multi-task convolutional network paired with a multi-objective hyperparameter optimization (HPO) framework that treats detection quality (PR-AUC) and localization quality (Dice) as simultaneous objectives. Using the Multiobjective Tree-structured Parzen Estimator (MOTPE) within a 10-fold nested cross-validation protocol (50 trials per outer fold, 500 unique configurations evaluated across 5 inner splits — 2 500 individual model trainings in total), DocVerify selects configurations from the Pareto-optimal front by minimizing Euclidean distance to the ideal point (1, 1), without any manual weighting. Evaluated on the FantasyID benchmark (Korshunov et al., 2025a), DocVerify achieves PR-AUC = 0.9921 ± 0.0058 and Dice = 0.856 ± 0.030 under nested cross-validation, and detection F1@0.5 = 0.969 ± 0.014 with localization F1 = 0.807 ± 0.096 on the blind test set (30 seeds). An ablation confirms that Pareto-optimized loss weighting provides a statistically significant Dice improvement over equal-weight joint training ($p = 1.3 \times 10^{-4}$, $d = 1.01$, large effect) while maintaining equivalent detection accuracy.

Keywords: identity document forensics, forgery detection, forgery localization, multi-task learning, multi-objective HPO, Pareto front, nested cross-validation, MOTPE

1. Introduction

Identity document fraud poses a growing threat to national security, financial integrity, and border control worldwide. The widespread availability of image-editing software has made it possible to produce convincing forgeries of passports, national identity cards, and driver’s licenses at low cost (Europol Innovation Lab, 2022). Automated forensic analysis of identity documents is therefore a critical component of modern digital identity verification pipelines.

Detection of forged identity documents is inherently a two-level problem. At the *document level*, a system must decide whether a presented document is genuine or has been tampered with (binary detection). At the *pixel level*, it must also identify and delineate which specific regions have been altered (forgery localization). Both objectives are operationally necessary.

Binary detection alone is insufficient for real-world forensic deployments. A positive detection decision provides no *actionable information* to the investigator: it does not reveal whether the fraudster altered the portrait photograph, falsified textual fields, or combined multiple manipulations. Without spatial evidence, a human expert cannot determine the nature of the fraud or build a legal case. Detection systems are also increasingly subject to *explainability requirements* under emerging AI regulations (European Parliament and Council of the European

Union, 2024): a binary “tampered” label cannot satisfy the obligation to provide a meaningful explanation of the automated decision. Conversely, localization alone does not yield a reliable document-level confidence score suitable for automated accept/reject decisions. A practical forensic system must address both tasks simultaneously.

The primary technical difficulty in joint detection and localization is that the two objectives are in *structural tension*: the classification head benefits from global image features, while the segmentation head requires spatially precise, pixel-level representations. Prior work either treats the two tasks independently (Guillaro et al., 2023) or collapses both losses into a scalar objective via a fixed weight λ (Caruana, 1997; Kendall et al., 2018). The choice of λ is arbitrary and dataset-dependent. A second challenge is the risk of over-fitting the model-selection criterion: HPO that maximizes PR-AUC may sacrifice Dice, and vice versa.

This paper introduces **DocVerify**, a multi-task forensic document analysis system that addresses both challenges. Our contributions are:

1. *Multi-objective Pareto HPO for multitask forensics.* Both PR-AUC and Dice are treated as simultaneous optimization objectives. The optimal configuration is selected from the Pareto front by minimizing the Euclidean distance to the ideal point (1, 1), requiring no manual task-weight tuning.
2. *DocVerify architecture.* An end-to-end multi-task CNN comprising a lightweight convolutional encoder coupled

*Corresponding author.

Email address: diazhernan@uniovi.es (Hernán Díaz Rodríguez)

to a U-Net decoder for pixel-level mask prediction and a classification head for document-level detection, trained jointly with a combined BCE+Dice segmentation loss.

3. *Rigorous nested cross-validation.* A 10-fold outer $\times$ 5-fold inner nested cross-validation scheme with 50 MOTPE trials per outer fold (2 500 HPO evaluations total). Final performance is estimated on a blind holdout using 30 independent seeds, with Wilcoxon signed-rank tests and Holm–Bonferroni correction.

The source code and experimental artifacts are publicly available at <https://github.com/HernanDiaz/IDVerify/>.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the DocVerify architecture and HPO framework. Section 4 presents experimental results. Section 5 discusses the findings and limitations. Section 6 concludes.

2. Related Work

2.1. Document and Image Forgery Detection

Deep learning has substantially advanced general-purpose image forensics: CNNs can detect subtle statistical fingerprints left by editing operations, often outperforming hand-crafted pipelines (Bappy et al., 2019; Wu et al., 2019). Cozzolino and Verdoliva (2020a) (Noiseprint) train a CNN via Siamese contrastive learning to extract camera-model noise residuals; forgeries appear as anomalies in this residual without requiring semantic understanding. Guillard et al. (2023) (TruFor) generalize this with a transformer fusion network (NoisePrint++) that jointly models RGB semantics and noise-pattern anomalies, achieving strong performance on general image forensics benchmarks with over 876 000 pre-training images.

Identity document forgery introduces unique challenges: forged regions are often small (a single text field or portrait), the structured background (guilloché patterns, UV-reactive inks) creates strong false negatives for general-purpose noise detectors, and document type diversity demands generalization beyond natural-image pre-training. Official evaluations on FantasyID confirm that general-purpose baselines (TruFor, MMFusion, FatFormer, UniFD) all achieve false-negative rates above 50% at a 10% false-positive threshold (Korshunov et al., 2025a), motivating domain-specific approaches.

Among systems trained exclusively on FantasyID, George and Marcel (2025) propose EdgeDoc, a lightweight CNN-Transformer encoder (EdgeNeXt) fed with the green channel and a NoisePrint residual, with a U-Net decoder optimized jointly for classification and localization. Yu et al. (2025) introduce Re-MTKD, a reinforcement-learning multi-teacher knowledge distillation framework using a ConvNeXt v2 encoder; the Sunlight team adapted this to identity documents using over 60 000 additional external images, winning both tracks of the DeepID 2025 Challenge (Korshunov et al., 2025b).

DocVerify differs from all prior document-forensics systems in three respects: (i) it systematically searches the task-weight λ_{mask} via multi-objective HPO rather than fixing it heuristically; (ii) it selects the final configuration via a Pareto-optimal criterion;

and (iii) it reports uncertainty through nested cross-validation, providing statistically rigorous performance estimates.

2.2. Multi-Task Learning and HPO

In multi-task learning (Caruana, 1997; Ruder, 2017), the dominant approach combines task losses via scalar scalarization (Marler and Arora, 2004) with fixed weights set before training. Kendall et al. (2018) propose learning these weights via homoscedastic uncertainty; Liu et al. (2019) introduce gradient normalization. All scalar approaches commit to a fixed trade-off before training begins.

Multi-objective HPO extends single-objective search to jointly optimize multiple metrics (Miettinen, 1999). The Multiobjective TPE (MOTPE) (Watanabe and Hutter, 2022) adapts the TPE surrogate (Bergstra et al., 2011) to the multi-objective setting, achieving competitive Pareto-front coverage with far fewer function evaluations than evolutionary methods such as NSGA-II (Deb et al., 2002)—critical when each evaluation requires training a deep network. To our knowledge, this is the first work to apply Pareto-based multi-objective HPO to joint identity-document forgery detection and localization within a nested cross-validation framework.

3. Methodology

3.1. Dataset

We train and evaluate DocVerify on FantasyID (Korshunov et al., 2025a), a publicly released benchmark for identity document forgery detection and localization developed by Ildi Research Institute for the DeepID 2025 Challenge (Korshunov et al., 2025b). FantasyID comprises approximately 3 284 labeled images of synthetic fantasy identity documents: *bonafide* (genuine) and *attack* (tampered via face swap or text inpainting).

Each image is accompanied by JSON annotations specifying bounding-box regions with `region_provenance`: `altered`. We convert these to binary segmentation masks at training resolution (224×224 pixels). We combine the FantasyID train and test partitions into a single pool and apply GroupShuffleSplit (grouping by document identity, `test_size = 0.15`, `random_state=42`) to obtain approximately 2 791 development images and 493 blind holdout images, ensuring all captures of the same document identity remain in the same partition.

3.2. Architecture

DocVerify is a multi-task encoder-decoder network for simultaneous document-level classification and pixel-level forgery localization (Fig. 1).

3.2.1. Encoder

The encoder is a six-block lightweight CNN inspired by Patel et al. (2023), with an increasing channel progression of $8 \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256$ filters. All layers use LeakyReLU ($\alpha = 0.2$) (Xu et al., 2015) and batch normalization (Ioffe and Szegedy, 2015). Blocks 1–4 use 3×3 convolutions; blocks 5–6 use 5×5 convolutions. Downsampling uses average pooling

PATEL CNN ENCODER

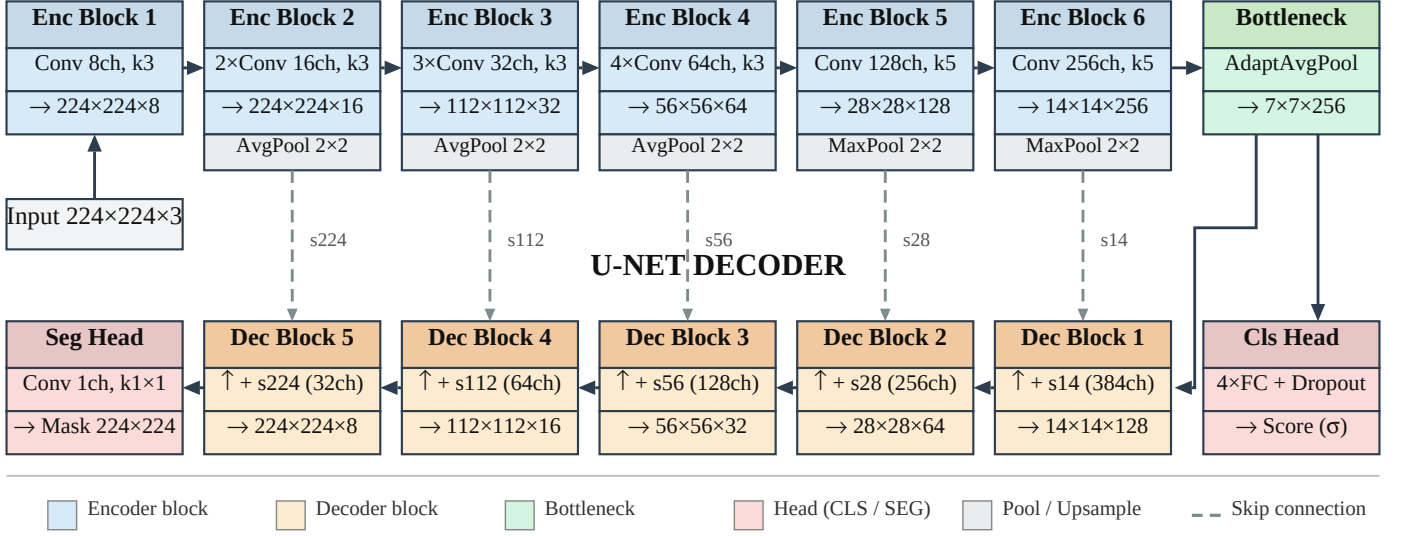


Figure 1: DocVerify architecture. The encoder maps a $3 \times 224 \times 224$ input to a $256 \times 7 \times 7$ bottleneck through six convolutional blocks ($8 \rightarrow 256$ channels). The U-Net decoder reconstructs the forgery mask via skip connections; the classification head outputs a document-level tampered/genuine score.

after blocks 2 and 3 (to preserve boundary information) and max pooling after blocks 4 and 5 (for translation invariance). Skip connections are extracted from each block prior to pooling.

3.2.2. Classification Head

Global average pooling is applied to the 7×7 bottleneck, yielding a 256-dimensional vector passed through four fully-connected layers ($256 \rightarrow 32 \rightarrow 16 \rightarrow 16 \rightarrow 1$) with dropout and LeakyReLU, producing a classification logit $\hat{y}_{\text{cls}}$.

3.2.3. Segmentation Head (U-Net Decoder)

A five-stage U-Net decoder (Ronneberger et al., 2015) reconstructs the spatial mask via bilinear upsampling and skip-connection concatenation. The channel width halves at each stage from C_{dec} (a tunable HPO parameter) down to $C_{\text{dec}}/16$. A final 1×1 convolution produces the segmentation logit $\hat{Y}_{\text{mask}}$.

3.3. Loss Functions

The combined training loss is:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{mask}} \mathcal{L}_{\text{seg}}, \quad (1)$$

where $\lambda_{\text{mask}} \in [0.5, 3.0]$ is included in the HPO search space. $\mathcal{L}_{\text{cls}} = \text{BCEWithLogits}(\hat{y}_{\text{cls}}, y_{\text{cls}})$. The segmentation loss combines BCE and Dice:

$$\mathcal{L}_{\text{seg}} = \text{BCE}(\hat{Y}, Y) + 1 - \frac{2 \sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon}, \quad (2)$$

as introduced for volumetric segmentation (Milletari et al., 2016) to address foreground/background imbalance in small forgery regions.

All models are trained with AdamW (Loshchilov and Hutter, 2019) and ReduceLROnPlateau (factor 0.5, patience 3). Mixed-precision training (FP16/FP32 via PyTorch AMP (Micikevicius et al., 2018)) is used throughout.

Table 1: HPO search space.

Parameter	Distribution	Range
η (learning rate)	log-uniform	$[5 \times 10^{-5}, 9 \times 10^{-4}]$
λ_{wd} (weight decay)	log-uniform	$[10^{-7}, 10^{-4}]$
p (dropout)	uniform	$[0.1, 0.4]$
C_{dec} (decoder channels)	categorical	$\{96, 128, 192, 256\}$
λ_{mask} (mask loss weight)	uniform	$[0.5, 3.0]$

3.4. Multi-Objective HPO

We optimize two complementary metrics: *PR-AUC* (area under the precision-recall curve, measuring document-level detection quality) and *Dice* ($2|P \cap G|/(|P| + |G|)$, measuring pixel-level mask overlap). Our HPO formulation simultaneously maximizes:

$$\max_{\theta \in \Theta} (\text{PR-AUC}(\theta), \text{Dice}(\theta)). \quad (3)$$

Table 1 defines the search space Θ .

We use MOTPE (Watanabe and Hutter, 2022) as implemented in Optuna (Akiba et al., 2019). After 50 trials on the inner splits, we select the configuration θ^* from the Pareto front $\mathcal{P}$ by minimizing the Euclidean distance to the ideal point:

$$\theta^* = \arg \min_{\theta \in \mathcal{P}} \sqrt{(1 - \text{PR-AUC})^2 + (1 - \text{Dice})^2}. \quad (4)$$

This criterion requires no user-specified weighting and assigns equal importance to detection and localization. Figure 2 illustrates the resulting objective space.

3.5. Nested Cross-Validation

To obtain unbiased estimates while keeping HPO blind to test data, we use a three-level nested CV scheme (Varma and Simon,

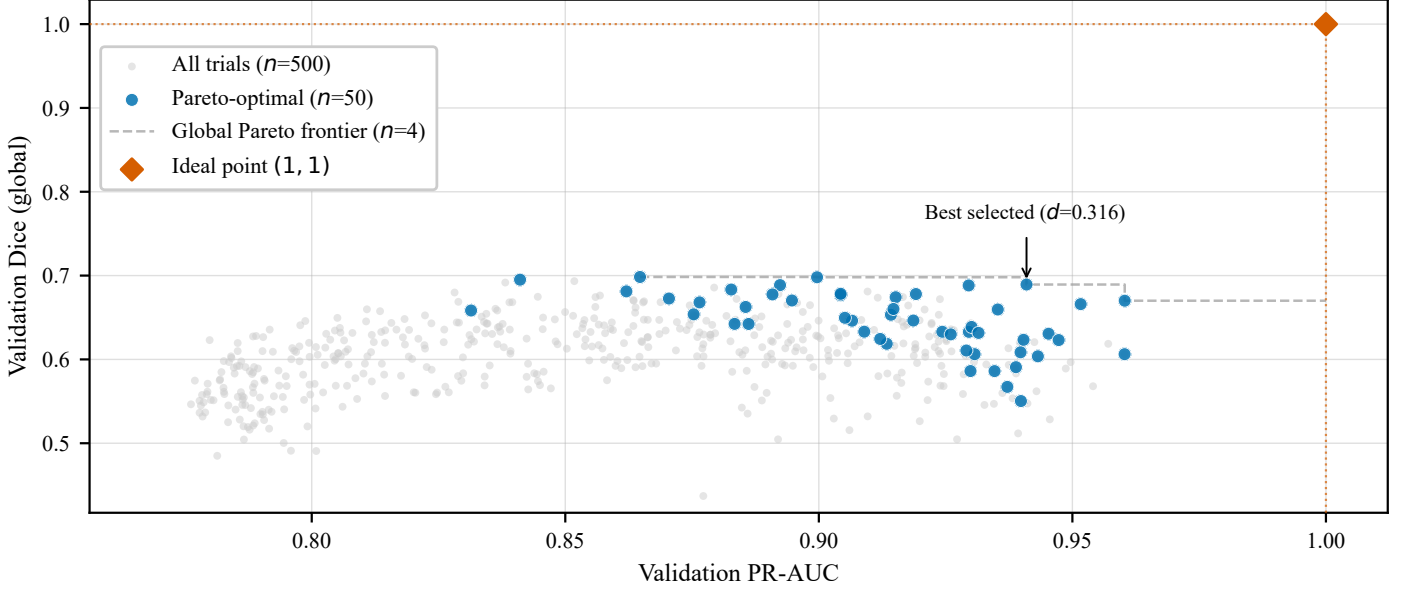


Figure 2: Validation objective space across all 500 HPO trials (10 folds $\times$ 50 MOTPE trials). Grey: evaluated; blue: Pareto-optimal; dashed: Pareto frontier. Red diamond: ideal point (1, 1); annotated point: selected configuration (distance $d = 0.316$).

2006). The outer loop uses 10 stratified folds; each fold serves once as the test set. For each outer fold, the remaining data is split into 5 inner folds on which 50 MOTPE trials are run (up to 15 epochs each, patience = 1). The optimal θ^* is then used to retrain from scratch on all outer-fold training data (up to 100 epochs, early stopping patience = 12). This design ensures HPO never sees test data: 50 MOTPE trials per outer fold yield **500 unique hyperparameter configurations** in total (10 outer folds $\times$ 50 trials); because each configuration is evaluated on 5 inner splits, this corresponds to $10 \times 50 \times 5 = 2\,500$ individual model trainings, plus 10 final models retrained on full outer-fold data.

4. Experiments and Results

4.1. Experimental Setup

All experiments are conducted on a single workstation running Windows 11, equipped with an AMD Ryzen 5 3400G (3.70 GHz), 32 GB DDR4-3200 RAM, and an NVIDIA RTX 5060 Ti (16 GB VRAM), using PyTorch 2.10.0 with CUDA 13.0 and Optuna 3.x. The full NCV pipeline required approximately 59 hours; images are cached in GPU memory per outer fold, reducing per-trial training time from ~ 12 to ~ 3.5 minutes.

4.2. Nested Cross-Validation Results

Table 2 reports per-fold test metrics. PR-AUC is consistently high across all folds (0.977–0.999, mean 0.9921 ± 0.0058). Dice shows slightly more fold-to-fold variability (0.800–0.909, mean 0.856 ± 0.030), reflecting the difficulty of pixel-level localization. The Pareto-selected λ_{mask} clusters between 2.3 and 2.7 in eight of ten folds; the two outlier folds (F5, F9) correspond to the two lowest Dice scores, confirming that λ_{mask} is a decisive driver of localization quality.

Table 2: Per-fold test results from 10-fold NCV. $\bar{w}_{\text{mask}}$: Pareto-selected loss weight.

Fold	PR-AUC	Dice	$\bar{w}_{\text{mask}}$
F1	0.9939	0.861	2.46
F2	0.9769	0.871	2.70
F3	0.9993	0.909	2.47
F4	0.9915	0.858	2.43
F5	0.9948	0.827	1.41
F6	0.9936	0.874	2.89
F7	0.9940	0.861	2.67
F8	0.9967	0.887	2.29
F9	0.9896	0.807	1.95
F10	0.9909	0.800	2.65
Mean	0.9921	0.856	2.39
Std	0.0058	0.030	0.43

4.3. Ablation Study

To isolate each design decision, we evaluate four variants on the blind test set using 30 independent seeds: *Multitask* (ours, joint detection + localization with Pareto-optimized λ_{mask}); *cls_only* (classification head only); *seg_only* (segmentation head only, detection by thresholding mean mask probability); and *unweighted_losses* (full multitask, $\lambda_{\text{cls}} = \lambda_{\text{mask}} = 0.5$, bypassing HPO). Table 3 and Fig. 3 summarize results.

Three findings stand out. First, joint multitask training is essential: *cls_only* produces near-zero Dice (0.002 ± 0.008). Second, a dedicated classification head is necessary: *seg_only* achieves competitive Dice (0.867) but PR-AUC drops to 0.728 ± 0.053 , confirming that the segmentation logit alone is an unreliable detection signal. Third, Pareto-optimized weights outperform equal weights specifically on localization:

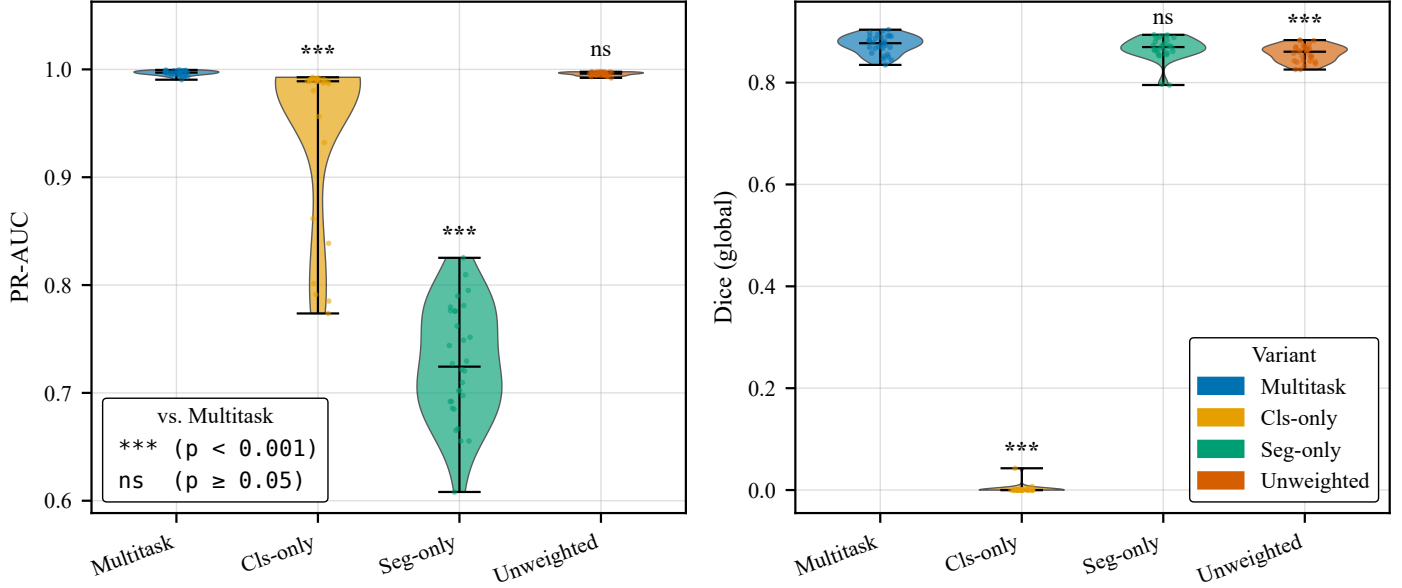


Figure 3: Ablation study: violin plots of PR-AUC (left) and Dice (right) for four variants over 30 seeds. Significance vs. Multitask: *** $p < 0.001$; ns $p \geq 0.05$ (Wilcoxon + Holm).

Table 3: Ablation study (blind test, 30 seeds). p : Wilcoxon + Holm; d : Cohen’s d . *** $p < 0.001$; ns $p \geq 0.05$.

Variant	PR-AUC		Dice			
	Mean ± Std	vs. Multitask	Mean ± Std	vs. Multitask	p_{holm}	d
Multitask (ours)	0.9967 ± 0.0021	—	0.875 ± 0.018	—	—	—
cls_only	0.9503 ± 0.0744	***	0.002 ± 0.008	***	1.3×10^{-8}	large
seg_only	0.7277 ± 0.0528	***	0.867 ± 0.023	ns	0.288	small
unweighted_losses	0.9958 ± 0.0016	ns	0.857 ± 0.016	***	1.3×10^{-4}	1.01

unweighted_losses matches Multitask on PR-AUC ($p = 0.083$) but shows significantly lower Dice ($p = 1.3 \times 10^{-4}$, $d = 1.01$, large effect), validating the HPO contribution.

4.4. Comparison with Related Systems

Table 4 compares DocVerify against the top participants of the DeepID 2025 Challenge on Track 1 (detection F1@0.5) and Track 2 (per-image localization F1), computed on our 15% holdout using 30 seeds.

DocVerify achieves F1@0.5 = 0.969 ± 0.014 on Track 1, surpassing all FantasyID-only systems including AG/EdgeDoc (+1.1 pp) and outperforming Incode despite its use of 100K proprietary images. On Track 2, DocVerify achieves F1_{loc} = 0.807 ± 0.096 , exceeding all four challenge finishers including the overall winner Sunlight (0.784). The high Track 2 variance (± 0.096) is driven by seeds where bonafide misclassifications generate false-positive mask pixels; F1 on attack images is stable (0.881 ± 0.015).

The comparison should be interpreted with caution. DocVerify was developed independently after the DeepID 2025 Challenge submission deadline had already closed; consequently, official test-set evaluation was not possible and we report results on our internal 15% holdout instead. Although both sets are drawn from the same FantasyID distribution, they are not

identical, making the comparison indicative rather than strictly controlled.

4.5. Qualitative Analysis

Figure 4 shows a representative output for a forged document with five altered regions. The predicted mask successfully localizes all five tampered fields — photograph, given name, date of birth, document number, and expiry date — with tight spatial alignment to the ground-truth alterations.

5. Discussion

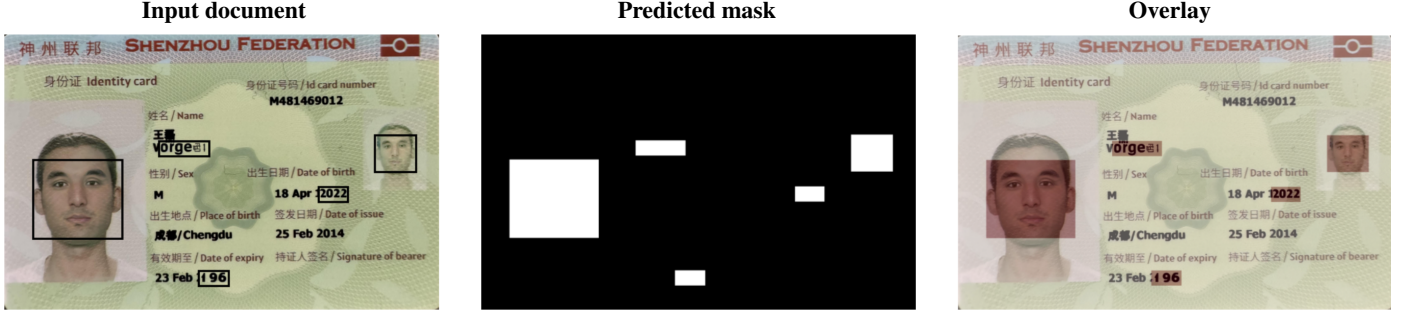
5.1. Analysis of Key Findings

The ablation reveals an asymmetry between the two tasks. The seg_only variant achieves competitive Dice (0.867 ± 0.023 , $p = 0.288$ vs. Multitask) but its PR-AUC collapses to 0.728 ± 0.053 : the U-Net decoder captures sufficient forgery evidence for pixel-level prediction yet cannot reliably aggregate it into a binary document-level decision. The result confirms that the shared encoder, regularized by the segmentation loss, learns spatially-aware features that strongly improve document-level detection.

Comparing Multitask against unweighted_losses isolates the contribution of HPO loss weighting. These variants differ

Table 4: Comparison with related systems. †: official challenge test set; DocVerify evaluated on internal holdout (see text).

System	F1 det (thr=0.5)	F1 loc	Training Data
Sunlight (Yu et al., 2025) (1st)†	0.991	0.784	60K+ external images
Incode (2nd)†	0.868	—	100K proprietary
AG/EdgeDoc (George and Marcel, 2025) (3rd)†	0.958	0.686	FantasyID only
UAM-Biometrics (4th)†	0.712	0.620	FantasyID only
TruFor (Guillaro et al., 2023) baseline†	0.807	0.590	876K pre-training
DocVerify (ours)	0.969 ± 0.014	0.807 ± 0.096	FantasyID only



Forged document — five altered regions localized

Figure 4: Qualitative result on a forged FantasyID sample. **Left:** input document. **Centre:** binary mask predicted by DocVerify. **Right:** predicted mask overlaid in red on the input. All five tampered fields are correctly localized.

only in whether λ_{mask} is set by Pareto HPO (mean 2.39 ± 0.43) or fixed at 0.5. Pareto-optimized weights yield significantly higher Dice ($p = 1.3 \times 10^{-4}$, $d = 1.01$, large effect) while maintaining equivalent PR-AUC, confirming that $\lambda_{\text{mask}} \approx 2.4$ is broadly optimal and that MOTPE surfaces this value reliably across folds.

5.2. Limitations

Single dataset.. All results are obtained on FantasyID, a synthetic dataset from a single research institute. Generalization to real-world identity documents, unseen document designs, or physical tampering attacks remains an open question.

Architecture.. The DocVerify encoder is a custom lightweight CNN trained from scratch. Transformer-based encoders (Doso-vitskiy et al., 2021) pre-trained on large-scale vision datasets may provide richer representations for cross-domain deployment.

Challenge evaluation gap.. The comparison in Table 4 compares our system on an internal holdout against challenge participants on the official test set. Although both sets are drawn from FantasyID, the results are indicative rather than strictly controlled.

Localization variance.. The high Track 2 standard deviation (± 0.096), driven by bonafide false-positive masks, represents a practical reliability concern. Targeted calibration strategies are a natural extension.

6. Conclusion

We presented DocVerify, a multi-task CNN for joint detection and localization of forged identity documents, paired with a multi-objective Pareto HPO framework that selects hyperparameter configurations by optimizing PR-AUC and Dice simultaneously without fixing their relative importance in advance.

Our main findings are: (i) joint multitask training with Pareto-optimized loss weights achieves significantly higher Dice than single-task variants and equal-weight joint training (blind test, 30 seeds, $p = 1.3 \times 10^{-4}$, $d = 1.01$) while maintaining competitive PR-AUC; (ii) under a rigorous 10-fold nested cross-validation protocol, DocVerify achieves PR-AUC = 0.9921 ± 0.0058 and Dice = 0.856 ± 0.030 , demonstrating consistent generalization across data partitions; and (iii) training exclusively on FantasyID without any external pre-training, DocVerify achieves results competitive with the top-3 entries of the DeepID 2025 Challenge, surpassing the winner on Track 2 localization (0.807 vs. 0.784) and all FantasyID-only systems on Track 1 detection.

Future work includes cross-domain evaluation on additional document datasets, integration of forgery-artifact-centric features (noise fingerprints, frequency anomalies (Cuzzolino and Verdoliva, 2020b)) for domain-invariant representations, and foundation model backbones (Muñoz-Haro et al., 2025) for richer pre-training.

Acknowledgments

This work was supported by the Instituto Nacional de Ciberseguridad (INCIBE, Spain’s National Cybersecurity Institute) and the INCIBE Chair in Cybersecurity and Artificial Intelligence

at the University of Oviedo, Spain. The author thanks Rafael González Quesada for his participation in the project, and the Idiap Research Institute for releasing the FantasyID dataset.

References

- Akiba, T., Sano, S., Yanase, T., Ohta, T., Koyama, M., 2019. Optuna: A next-generation hyperparameter optimization framework, in: Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, pp. 2623–2631. doi:[10.1145/3292500.3330701](https://doi.org/10.1145/3292500.3330701).
- Bappy, J.H., Simons, C., Nataraj, L., Manjunath, B.S., Roy-Chowdhury, A.K., 2019. Hybrid LSTM and encoder–decoder architecture for detection of image forgeries, pp. 3286–3300. doi:[10.1109/TIP.2019.2895466](https://doi.org/10.1109/TIP.2019.2895466).
- Bergstra, J., Bardenet, R., Bengio, Y., Kégl, B., 2011. Algorithms for hyperparameter optimization, in: Advances in Neural Information Processing Systems (NeurIPS).
- Caruana, R., 1997. Multitask learning. Machine Learning 28, 41–75. doi:[10.1023/A:1007379606734](https://doi.org/10.1023/A:1007379606734).
- Cohen, J., 1988. Statistical Power Analysis for the Behavioral Sciences. 2nd ed., Lawrence Erlbaum Associates.
- Cozzolino, D., Verdoliva, L., 2020a. Noiseprint: A CNN-based camera model fingerprint. IEEE Transactions on Information Forensics and Security 15, 144–159. doi:[10.1109/TIFS.2019.2916364](https://doi.org/10.1109/TIFS.2019.2916364).
- Cozzolino, D., Verdoliva, L., 2020b. Noiseprint: A CNN-based camera model fingerprint. IEEE Transactions on Information Forensics and Security 15, 144–159. doi:[10.1109/TIFS.2019.2916364](https://doi.org/10.1109/TIFS.2019.2916364).
- Deb, K., Pratap, A., Agarwal, S., Meyarivan, T., 2002. A fast and elitist multi-objective genetic algorithm: NSGA-II. IEEE Transactions on Evolutionary Computation 6, 182–197. doi:[10.1109/4235.996017](https://doi.org/10.1109/4235.996017).
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., Houlsby, N., 2021. An image is worth 16x16 words: Transformers for image recognition at scale, in: International Conference on Learning Representations (ICLR). URL: <https://openreview.net/forum?id=YicbFdNTTy>, arXiv:2010.11929.
- European Parliament, Council of the European Union, 2024. Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). Official Journal of the European Union, OJ L 2024/1689. URL: <https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>, published 12 July 2024; entered into force 1 August 2024. Article 13: Transparency and provision of information to deployers.
- Europol Innovation Lab, 2022. Facing Reality? Law Enforcement and the Challenge of Deepfakes. Observatory Report. Europol. The Hague, Netherlands. URL: <https://www.europol.europa.eu/publications-events/publications/facing-reality-law-enforcement-and-challenge-of-deepfakes>.
- George, A., Marcel, S., 2025. EdgeDoc: Hybrid CNN-transformer model for accurate forgery detection and localization in ID documents. URL: <https://arxiv.org/abs/2508.16284>, arXiv:2508.16284. idiap Research Report RR-08-2025; 3rd place DeepID 2025 Challenge.
- Guillaro, F., Cozzolino, D., Sud, A., Dufour, N., Verdoliva, L., 2023. TruFlo: Leveraging all-round clues for trustworthy image forgery detection and localization, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 19782–19792. doi:[10.1109/CVPR52729.2023.01895](https://doi.org/10.1109/CVPR52729.2023.01895).
- Ioffe, S., Szegedy, C., 2015. Batch normalization: Accelerating deep network training by reducing internal covariate shift, in: Proceedings of the 32nd International Conference on Machine Learning (ICML), pp. 448–456.
- Kendall, A., Gal, Y., Cipolla, R., 2018. Multi-task learning using uncertainty to weigh losses for scene geometry and semantics, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 7482–7491. doi:[10.1109/CVPR.2018.00781](https://doi.org/10.1109/CVPR.2018.00781).
- Korshunov, P., Mohammadi, A., Vidit, V., Ecabert, C., Marcel, S., 2025a. FantasyID: A dataset for detecting digital manipulations of ID-documents, in: Proceedings of the IEEE International Joint Conference on Biometrics (IJCBI), Osaka, Japan. URL: <https://arxiv.org/abs/2507.20808>, arXiv:2507.20808.
- Korshunov, P., Vidit, V., Mohammadi, A., Ecabert, C., Shamoska, N., Marcel, S., et al., 2025b. DeepID: Challenge of detecting synthetic manipulations in ID documents, in: Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW), pp. 510–519. URL: https://openaccess.thecvf.com/content/ICCV2025W/DeepID/html/Korshunov_DeepID_Challenge_of_Detecting_Synthetic_Manipulations_in_ID_Documents_ICCVW_2025_paper.html.
- Liu, S., Johns, E., Davison, A.J., 2019. End-to-end multi-task learning with attention, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 1871–1880. doi:[10.1109/CVPR.2019.00197](https://doi.org/10.1109/CVPR.2019.00197).
- Loshchilov, I., Hutter, F., 2019. Decoupled weight decay regularization, in: International Conference on Learning Representations (ICLR).
- Marler, R.T., Arora, J.S., 2004. Survey of multi-objective optimization methods for engineering. Structural and Multidisciplinary Optimization 26, 369–395. doi:[10.1007/s00158-003-0368-6](https://doi.org/10.1007/s00158-003-0368-6).
- Micikevicius, P., Narang, S., Alben, J., Diamos, G., Elsen, E., Garcia, D., Ginsburg, B., Houston, M., Kuchaiev, O., Venkatesh, G., Wu, H., 2018. Mixed precision training, in: International Conference on Learning Representations (ICLR).
- Miettinen, K., 1999. Nonlinear Multiobjective Optimization. Kluwer Academic Publishers. doi:[10.1007/978-1-4615-5563-6](https://doi.org/10.1007/978-1-4615-5563-6).
- Milletari, F., Navab, N., Ahmadi, S.A., 2016. V-Net: Fully convolutional neural networks for volumetric medical image segmentation, in: Proceedings of the 4th International Conference on 3D Vision (3DV), pp. 565–571. doi:[10.1109/3DV.2016.79](https://doi.org/10.1109/3DV.2016.79).
- Muñoz-Haro, J., Tolosana, R., Vera-Rodriguez, R., Morales, A., Fierrez, J., 2025. FakeIDet: Exploring patches for privacy-preserving fake ID detection, in: Proceedings of the IEEE International Joint Conference on Biometrics (IJCBI). URL: <https://arxiv.org/abs/2504.07761>, arXiv:2504.07761.
- Patel, Y., Tanwar, S., Bhattacharya, P., Gupta, R., Alsuwian, T., Davidson, I.E., Mazibuko, T.F., 2023. An improved dense CNN architecture for deepfake image detection. IEEE Access 11, 22081–22095. doi:[10.1109/ACCESS.2023.3251394](https://doi.org/10.1109/ACCESS.2023.3251394). open access, CC BY-NC-ND 4.0.
- Ronneberger, O., Fischer, P., Brox, T., 2015. U-Net: Convolutional networks for biomedical image segmentation, in: Medical Image Computing and Computer-Assisted Intervention (MICCAI), pp. 234–241. doi:[10.1007/978-3-319-24574-4_28](https://doi.org/10.1007/978-3-319-24574-4_28).
- Ruder, S., 2017. An overview of multi-task learning in deep neural networks. arXiv preprint arXiv:1706.05098.
- Varma, S., Simon, R., 2006. Bias in error estimation when using cross-validation for model selection. BMC Bioinformatics 7, 91. doi:[10.1186/1471-2105-7-91](https://doi.org/10.1186/1471-2105-7-91).
- Watanabe, S., Hutter, F., 2022. Multiobjective tree-structured parzen estimator, pp. 1–31. doi:[10.1613/jair.1.13188](https://doi.org/10.1613/jair.1.13188).
- Wu, Y., AbdAlmageed, W., Natarajan, P., 2019. ManTra-Net: Manipulation tracing network for detection and localization of image forgeries with anomalous features, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 9543–9552. doi:[10.1109/CVPR.2019.00977](https://doi.org/10.1109/CVPR.2019.00977).
- Xu, B., Wang, N., Chen, T., Li, M., 2015. Empirical evaluation of rectified activations in convolutional network. arXiv preprint arXiv:1505.00853.
- Yu, Z., Ni, J., Zhang, J., Deng, H., Lin, Y., 2025. Reinforced multi-teacher knowledge distillation for efficient general image forgery detection and localization, in: Proceedings of the AAAI Conference on Artificial Intelligence. URL: <https://arxiv.org/abs/2504.05224>, arXiv:2504.05224. oral presentation; base method of Sunlight (1st place, DeepID 2025).